

Demand and innovation: Schmookler re-examined *

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Schmookler's "demand-pull" hypothesis, implying that innovation is a function of market demand, has been widely accepted. Re-reading Schmookler's *Invention and Economic Growth* we discover a statistical inadequacy. Re-estimating his cross-section results in a different way, we find somewhat lower coefficients. Moreover, the direction of causality appears to be far from obvious. An independent cross-sectional test of the Schmookler hypothesis in the Netherlands, based on R&D instead of patent data, shows similar results: the relationship between indicators of demand and innovation turns out to be generally weaker (but in most cases still significant), and the results suggest that there is a simultaneous relationship between demand and innovation. The latter may be explained by the concept of the "innovation multiplier", which has implications for macro-economic modelling.

1. Re-reading Schmookler

Schmookler's [13] finding of a strong relationship between investment in capital goods users industries and patent applications by capital goods producing sectors has been generally accepted by economists as evidence that patenting is a function of effective demand ("demand pull" hypothesis). Quite a number of case studies have followed Schmookler's seminal work, but there have been

only few quantitative tests of his hypothesis (e.g., [12,15]).

In a detailed survey of post-Schmooklerian case studies on "demand-pull", Mowery and Rosenberg [10] gave plausible arguments against a naive acceptance and over-interpretation of "demand-pull". They argue that "demand-pull" has often been interpreted in terms of "need-pull", and that such a broad concept leaves little chance for falsification. Walsh [14] and Achilladelis et al. [1] have found additional evidence against a one-sided interpretation of "demand-pull".

While sympathizing with such points of critique, we found additional reasons for doubts about the "demand pull" hypothesis when re-reading Schmookler's original work [13]. Our doubts relate to the strength of the relationship between investment and patenting as well as to the direction of causality.

Let us begin with the latter. Wyatt's [15] Granger causality test on Schmookler's time series confirmed that investment is indeed lagging behind patenting. Does such a finding allow the conclusion that invention is a function of effective demand? There are at least two reasons for doubts.

First, in interpreting Schmookler's data, one should remind oneself that patents are a *result* of inventive activity; that is, an increase or decrease of patenting activity may lag several years behind increases or decreases in inventive effort. Hence a lag of patenting behind investment does not necessarily prove that fluctuations in inventive activity *follow* fluctuations in investment.

Second, there may be a lag between inventions and patent applications, as Schmookler himself

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[13, p. 24] points out:

Since the patent law of the United States permits up to one year of public use prior to filing of an application without destroying the inventor's right to a patent, and since a captive inventor's own firm usually provides a ready testing ground, it is not uncommon for corporations to test their inventions commercially without filing applications and then to refrain from filing on inventions which prove unprofitable.

Apart from rendering questionable the conventional wisdom that patents are indicators of invention (as opposed to innovation), the above quotation raises further doubt about whether Schmookler's and Wyatt's results indeed support a unidirectional causality from demand to inventive effort as implied in Schmookler's demand-pull hypothesis. An opposite direction of causality ("technology-push") appears to be at least equally plausible.

There are also theoretical reasons to think so. It is of course true that strong demand is positive for patenting because it increases returns to inventive activity. However, it can also be argued that inventive and innovative activity itself increases demand ("technology-push"), owing to the multiplier and accelerator effects associated with the launching of innovations; that is, inventive and innovative activity is supposed to require investment in R&D, know-how and capacity build-up. The latter may be conceived of as a positive function of the degree of radicalness of the innovation and its degree of market success (diffusion). Following this line of reasoning, "demand-pull" and "technology-push" effects might be conceived of as complementary rather than mutually exclusive.

It can be added here that in the recent neo-Schumpeterian literature it has been suggested that the relative weight of "demand-pull" and "technology-push" may vary with the stage of the "life-cycle" of an industry and with the type of innovation; that is, "technology-push" may be more important for important innovative breakthroughs in the beginning of the life cycle, while "demand-pull" may be more important for subsequent ("secondary") innovations (see, e.g., Walsh [14] and Achilladelis et al. [1]; for a discussion in the context of long waves see, e.g., Free-

man et al. [5] and Kleinknecht [8]). However, this point cannot be pursued here, since it is plausible that Schmookler's patent data (as well as our data from the Netherlands, see below) are mainly indicators of "secondary" innovations.

In addition to doubts about the direction of causality, we also discovered reasons to question the strength of the cross-sectional relationship between demand and patenting in Schmookler's work. For example, Schmookler regresses logs of patent applications (from 1940–42) on logs of investment (in 1939), finding an R^2 of .918. Regressing patent applications for the period 1948–50 on investment in 1947, and trying out various leads and lags as well, he obtains very similar results [13, p. 147].

There is a methodological problem behind these regressions since Schmookler took the logs of the *absolute* amount of investments and patents per sector. As he concedes himself, "High correlations between two variables are often the result of their joint dependence on a third variable. The size of industry may be the third variable in this case. Thus it may be argued that investment and invention will tend to vary directly with the size of industry" [13, p. 144]. This consideration appears realistic, since Schmookler's industry sectors differ tremendously in size [13, p. 146].

He tries to deal with this objection by including in his regression equation an indicator of sector size (i.e., employment per sector) as a second explanatory variable, concluding that this second variable proved to be insignificant [13, p. 145]. This is an invalid inference, however, for the size variable is highly correlated with investment and patenting;¹ that is, there is a multicollinearity problem.

Schmookler makes still another attempt to correct for the influence of size differences by dividing patents and investments by numbers of employees per sector. Regressions on the resulting ratios are similarly strong [13, p. 145]. However, this procedure may also be unreliable. As Kuh and Meyer [9] have shown, under certain conditions division of both sides of the regression equa-

¹ Correlating investment, patenting and employment in Schmookler's data set, we obtain coefficients of correlation ranging from 0.845 to 0.924.

tion by the same variable may lead to spuriously high coefficients.²

If Schmookler's correlations between investment and patenting were indeed caused by the size variable, this would have serious consequences. In fact, his "explanation" of patenting by investment would then come down to the trivial statement that large sectors (simply because of being large) have more investments and patents than smaller sectors and vice versa. In principle, this would even be consistent with the proposition that innovation is *exogenous*: if patents came about randomly just like wind and rain, it would not be surprising if we discovered that large trees catch more wind than smaller ones. Hence the proposition that "The amount of invention is governed by the extent of the market" [13, p. 137] would imply a meaningless concept of "demand-pull". There is therefore a need to check for the validity of his results.

2. Alternative estimates on Schmookler's data

It follows from the above that, for testing of the "demand-pull" hypothesis, measures of patenting and demand which are independent of sector sizes ought to be used. Principally, we see two theoretically plausible ways of doing this. The first would be to use *partial* correlations, relating (absolute) investment and patents to each other, the influence of sector size being kept constant; the second possibility would consist of using *growth rates* of investment and of patenting, arguing that the rise or decline of patenting activity is somehow related to *changes in* (as opposed to *size of*) demand. The growth rate method has the advantage that it can be applied to Schmookler's patent and value added data (1899–1947, see [13, pp. 153–7]) as well as to his investment and patent data (1936–1950, see [13, pp. 142–4]). Partial correlations can be applied to the latter only, since these cover a measure of sector size. Both methods appear to be theoretically consistent with the de-

Table 1

Partial cross-section correlations between (absolute) investment and patenting, the influence of sector size being kept constant^a

1939 investment and patents applied for in:			1947 investment and patents applied for in:		
	(1)	(2) ^b		(1)	(2) ^b
1936–38	0.714	(0.939)	1945–47	0.610	(0.922)
1937–39	0.576	(0.938)	1946–48	0.714	(0.939)
1938–40	0.783	(0.948)	1947–49	0.703	(0.945)
1939–41	0.819	(0.953)	1948–50	0.709	(0.951)
1940–42	0.838	(0.958)	1949–50	0.648	(0.952)
1941–43	0.810	(0.951)	1950	0.550	(0.940)

^a Calculations are based on Schmookler's data [13, pp. 140–3].

^b Correlations *without* correction for size differences are given in parentheses; the latter are calculated by taking square roots from Schmookler's R^2 [13, p. 147].

mand-pull hypothesis. The results are documented in tables 1 and 2. In both tables we follow Schmookler in taking logs of investment and patents. Besides testing the demand-pull hypothe-

Table 2

Summary of bivariate regression results (*t*-values in parentheses)

(a) Leads of investment over patents ("demand-pull")

- (1) $\log \text{GLP41/49} = -2.64 + 0.4575 \log \text{I39/47};$
(-0.822) (0.799)
 $R^2 = 0.000, N = 20$
- (2) $\log \text{GLP42/49} = 4.16 - 0.5515 \log \text{I39/47};$
(0.606) (-0.452)
 $R^2 = 0.000, N = 20$
- (3) $\log \text{GLP46/49} = 3.10 - 0.0412 \log \text{I39/47};$
(0.087) (-0.007)
 $R^2 = 0.000, N = 20$

(b) Leads of patents over investment ("technology-push")

- (4) $\log \text{I39/47} = 5.65 + 0.0550 \log \text{GLP37/40};$
(62.096) (1.705)
 $R^2 = 0.091, N = 20$
- (5) $\log \text{I39/47} = 5.69 + 0.0660 \log \text{GLP37/41};$
(54.961) (1.584)
 $R^2 = 0.074, N = 20$
- (6) $\log \text{I39/47} = 5.77 + 0.0735 \log \text{GLP37/42};$
(46.920) (1.921)
 $R^2 = 0.124, N = 20$

Notes

GLP.../... = average annual growth rate of logs of patents from the year 19... to the year 19....

I39/47 = average annual growth rate of logs of investment from 1939 to 1947.

Source: Schmookler [13, pp. 140–3].

² Kuh and Meyer [9] have shown that ratio correlations differ from partial correlations between the numerators when the latter are non-linear homogeneous functions of the deflator. This is the case in Schmookler's data. Comparison between our partial correlations to be presented further below, and Schmookler's ratio regressions [13, p. 145] confirms this.

sis, we also try out leads of patents over investment ("technology-push").

The partial correlations in table 1 (column 1) are considerably lower than correlations without correction for the influence of sector size (column 2). This confirms our above argument that the high R^2 s reported by Schmookler are in part due to size differences between sectors. Nonetheless, the remaining correlations are still strong.

It is interesting to compare in table 1 the coefficients for different leading and lagging periods. Commenting on his regressions for the periods as given in table 1 (column 2), Schmookler concludes that "the coefficients tend to rise from the start and peak about two years *after* the investment occurred. This suggests that the main direct causal force is from investment to invention" [13, p. 147] (Schmookler's italics). Judging from our partial correlations (column 1), Schmookler's interpretation is less obvious, in particular for correlations of patents with 1947 investment. Nonetheless, apart from the causality issue, the hypothesis of a correlation between investment and patenting can be maintained.

In interpreting both tables, it should be kept in mind that observations are based on 20 sectors only, and in the case of table 2 (notably in equations 1 and 2), the data did not allow us to handle more adequate time lags. Having these qualifications in mind, it can be concluded that the demand-pull hypothesis receives no support from the regressions in table 2. Only in the technology-push case are t values significant at the 95 percent level.

As to tables 1 and 2, one has to remember as another important qualification that Schmookler's above-tested data may have been disturbed by World War II. It is therefore interesting also to consider his data for earlier periods. In a separate section, Schmookler analyses patent applications and value added for selected years in 16 industries in the period 1899–1947, finding equally strong correlations [13, pp. 151–162]. In principle, our above methodological criticism also applies to these estimates.

Schmookler's data allow us to examine the relationship of patent applications growth and value-added growth for specific periods, including leads of patenting on value added ("technology-push"). In order to economize on space, we restrict ourselves to documenting in table 3 only coefficients

of correlation, indicating with an asterisk coefficients which are significantly different from zero at the 95 percent level.

Following Schmookler, we only document the coefficients calculated on the *logs* of patenting and value added (coefficients on the original values have also been calculated and proved to be somewhat lower, in general). Moreover, table 3 is restricted to correlations between patenting and value added in *current* prices, which differ little from estimates using *constant* prices.

In interpreting table 3, one needs to keep in mind that the cross-section coefficients are calculated from relatively highly aggregated data, distinguishing only 16 sectors. Hence the coefficients may be very sensitive to outliers. We therefore checked the scattergrams of all correlations in table 3, adding a question mark to those coefficients which were obviously strongly influenced by one or two outliers.

Table 3 is difficult to interpret. First of all, looking at coefficients for leading and lagging periods, we find only a few significant coefficients besides a number of insignificant or even negative ones; that is, there is no particular support for either "demand-pull" or "technology-push". Most of the positive coefficients can be observed when periods overlap. But even in these cases some insignificant and negative coefficients forbid the drawing of strong conclusions.

While part of the low coefficients may be explained by disturbances caused by the two world wars and by the extraordinary depression of 1929–32, there is no explanation for the low coefficients in the first decade of this century. On the other hand, during the 1920s we find quite good coefficients, also when regarding longer periods.³ In any case as far as table 3 supports the hypothesis of mutual dependence of patenting and demand, the relationship is definitely weaker than suggested by Schmookler.

3. A test of the Schmookler hypothesis in Dutch industry

It should be noted that the hypothesis of a correlation between investment (by patent "users")

³ For example, correlating the average annual growth rates of patenting (1921–29) with growth rates of value added (1919–29) we obtain an R of 0.70.

Table 3
Correlations between value added growth (current prices) and patent applications growth, selected years (1899–1947)

Average annual growth of logs of: Patent applications from 19... to 19...															
	01-06	06-11	11-16	16-21	21-23	23-25	25-27	27-29	29-31	31-33	33-35	35-37	37-39	39-41	41-49
Value added															
1899-1904	-0.05	-0.34?	0.07												
1904-1909	-0.40?	-0.52?	0.08												
1909-1914	0.00	0.47*	0.71*	-0.39											
1914-1919		-0.24?	0.44*	-0.16	0.11										
1919-1921			0.32	-0.52*	0.32	0.59?									
1921-1923			0.32	0.23	0.62*	-0.29	0.09								
1923-1925				0.19	-0.28	0.60*	0.46*	0.36							
1925-1927					-0.20	0.15	0.06	-0.09							
1927-1929						0.49*	0.59*	0.39	0.37	0.16					
1929-1931								0.46*	0.26	0.50*	0.36				
1931-1933								0.68*	-0.11	0.38	-0.27	-0.13			
1933-1935									0.28?	-0.30?	0.00	0.02	-0.55?		
1935-1937										-0.03	-0.62*	0.10	0.54*	0.42?	
1937-1939											-0.39?	0.02	0.83*	0.86?	-0.12
1939-1947												0.13	-0.24	-0.43?	-0.23

Notes

? = coefficient is less reliable because of one or two outliers.

* = coefficient is significantly different from zero at 95% level.

Calculations are based on Tables 24 and 25 in Schmookler [13, pp. 152–3].

In the case of patent applications, average annual growth rates are based on the middle years (taking arithmetic means) of the three-year periods as given by Schmookler.

and patent applications (by patent “producers”) is not the only possible way in which the demand-pull hypothesis can be specified. Schmookler himself tests data from Scherer [11] on patents granted by sector of origin and indicators of demand *in the same sectors*. Of course, when examining the demand-pull hypothesis in investment goods-producing industries, it is obvious that investment in investment goods-using industries is a good indicator of demand. However, when examining the demand-pull hypothesis in a more general setting than in the investment goods market, sales figures of the patent “producing” industries are the better indicator of demand.

Regressing Scherer’s patent data on such indicators of demand (i.e., sales, assets and employment), Schmookler finds somewhat lower significance levels (R^2 s ranging from 0.429 to 0.442, see [13, p. 171]). It is not difficult to guess that his lower significance levels also have to do with less dramatic differences in sector sizes in this case,⁴ and that probably the same holds for the Scherer [12] study.⁵

In our cross-sectional test on the relationship between demand and innovation in the Netherlands, we also use indicators of innovation and of demand relating to the *same* sector. In order to avoid a size bias, we work with innovativeness related to sector size and with growth rates of sales and investment. Our innovation data stem from a large-scale mailing survey on innovation in Dutch manufacturing industry (see Kleinknecht [7]). The data on sales and investment come from the Dutch Central Statistical Office [3,4]. Our innovation data refer to the year 1983. The investment and sales data cover the periods 1980–86 and 1981–85 respectively, and will therefore allow us to test the “demand-pull” as well as the “technology-push” hypothesis.

While our Dutch data have the disadvantage of referring to one year only (1983), they have two advantages as compared with those by Schmookler. First the innovation data are available at the firm level and, therefore, we can handle a much finer level of disaggregation, distinguishing 46 instead of 20 (or 16) sectors in Schmookler’s work; second, our data cover the entire manufacturing industry of the Netherlands, while Schmookler’s data were restricted to a subset of U.S. manufacturing.

It should be added that our innovation data refer to man-hours spent on R&D. The latter are an input indicator of the innovation process and are therefore rather different from Schmookler’s patents, which are an output indicator. On the other hand, as the reliability of patents as an innovation output indicator has been frequently discussed (for a recent survey, see e.g., Basberg [2]) it is interesting to see whether the results obtained from patent data also hold when applying a different innovation indicator.

In table 4 we summarize regressions between growth rates of sales and investment and R&D intensity (RDI). The latter is defined as the total R&D man-years in a sector in 1983 divided by total labour-force (man-years) per sector. We distinguish between the “demand-pull” case (explaining R&D by sales) and the “technology-push” case (explaining sales by R&D). In the latter case, following the above-mentioned multiplier/accelerator principle, we also check for the relationship between R&D and investment. In order to economize on space we do not document a number of regressions which have been run in order to check the robustness of the results. For example, we ran logarithmic regressions, the results of which differed only very little from those presented here.

The parameters in table 4 confirm that there is a statistically significant relationship between R&D and demand growth, although the regressions on growth rates generally show a weaker association than would probably be obtained by levels regressions (see, as an illustration, the partial correlations in table 1). In table 4 again, evidence of “technology-push” is approximately equally strong as evidence of “demand-pull”. It is remarkable to see that the R&D variable is significantly correlated not only with a “direct” indicator of demand (i.e., sales) but also with investment. This suggests that, in spite of weak growth and over-capacities in Dutch industry, the above-

⁴ Depending on whether we take sales, assets or employment as an indicator of sector size, the difference between the smallest and the largest sector varies between 1:10 and 1:17 [13, p. 169].

⁵ Taking Schmookler’s employment data as a size indicator [13, pp. 140–3], the difference between the smallest and the largest sector amounts to 1:157 in the 1939 case and to 1:136 in the 1947 data. In Scherer’s [12] data, however, the difference between the smallest and the largest sector is only 1:72. His regression estimates can therefore be expected to be less seriously upward biased than those by Schmookler.

mentioned multiplier/accelerator mechanism seems to work.

Our innovation data allow us to examine the relationship between innovation and demand in still another way, using data on sales growth which have been reported in our mailing survey. The survey data do not allow handling time lags between innovation and sales, for all survey data relate to 1983. But the data do allow us to check the relationship at the firm level. Since data on sales growth are not available at a continuous base, we have to use cross-tabulation, and apply a χ^2 test. The cross-tabulation in table 5 shows that there is a positive relationship between sales and

Table 4

Summary of cross-sectional regressions between demand and innovation in 46 sectors of Dutch manufacturing industry (*t*-values in parentheses)

(a) "Demand-pull": demand (i.e. sectoral differences in 1981–83 and 1982–83 growth rates of sales) causing sectoral differences in 1983 innovative efforts (RDI = R&D man-years as a percentage of total manpower per sector)

- (1) $RDI = 2.31 + 0.1131 \text{ SALES8183}$; adj. $R^2 = 0.113$; $n = 46$
(5.411) (2.596)
(2) $RDI = 2.58 + 0.1352 \text{ SALES8283}$; adj. $R^2 = 0.144$; $n = 46$
(6.872) (2.929)

(b) "Technology-push": sectoral differences in 1983 innovative effort (RDI) causing sectoral differences in demand growth (i.e., 1983–84 and 1983–85 growth rates of sales)

- (3) $\text{SALES8384} = 2.38 + 1.7483 \text{ RDI}$; adj. $R^2 = 0.092$; $n = 46$
(0.999) (2.341)
(4) $\text{SALES8385} = 9.05 + 2.1845 \text{ RDI}$; adj. $R^2 = 0.106$; $n = 46$
(3.239) (2.491)

Note: equations (3) and (4) each exclude one outlier. Inclusion of the outlier would increase the R^2 in equation (3) from 0.092 to 0.405 and in equation (4) from 0.106 to 0.166.

(c) "Technology-push" (accelerator approach): sectoral differences in innovative effort (RDI) in 1983 causing sectoral differences in investment growth (1983–85 and 1983–86)

- (5) $\text{INVEST8385} = 35.85 + 11.9983 \text{ RDI}$;
(2.368) (2.518)
adj. $R^2 = 0.108$; $n = 46$
(6) $\text{INVEST8386} = 49.23 + 13.0438 \text{ RDI}$;
(3.476) (2.927)
adj. $R^2 = 0.146$; $n = 46$

Note: equations (5) and (6) each exclude one outlier. Inclusion of the outlier would bring down the R^2 in equation (5) from 0.108 to 0.044 (*t*-value of coefficient: 1.7515), and in equation (6) from 0.146 to 0.020 (*t*-value: 0.9676).

Table 5

Cross-tabulation of R&D intensity and sales growth in 1842 Dutch manufacturing firms

As compared with 1982, your 1983 sales were:	R&D-intensity						Row totals
	No R&D	0 to < 2%	2 to < 4%	4 to < 6%	6 to < 8%	$\geq 8\%$	
More than 20% lower	30	18	14	8	2	7	79
10–20% lower	40	40	22	5	5	2	114
0–10% lower	77	93	31	11	12	11	235
almost the same	152	149	71	26	18	23	439
0–10% higher	155	161	96	41	24	28	505
10–20% higher	69	73	41	34	16	36	269
more than 20% higher	46	48	46	18	9	31	198
Column totals	569	582	321	143	86	138	1839

Notes

Definition of R&D intensity: R&D man-years as a percentage of total manpower (man-years) of a firm.

There are 3 missing values among 1842 questionnaires returned.

The χ^2 value of 99.03 is significant at a 99.9% level (assuming 30 degrees of freedom).

Source: Kleinknecht [7].

innovation at the firm level. This confirms our above findings at the sectoral level.

4. Conclusions

The above exercises confirm that the cross-section regressions as reported by Schmookler are indeed spuriously high because of his inadequate correction for the influence of differences in sector sizes. Our application of partial correlations (table 1) confirms that Schmookler's patents and investments are somewhat weaker (but still significantly) correlated. As compared with partial correlations, regressions and correlations on growth rates (tables 2 and 3) give weaker (but in many cases still significant) evidence of a positive relationship. However, one has to bear in mind that regressions on Schmookler's data are based on a

relatively low number of observations ($n = 20$ and $n = 16$). From our Dutch test, it is interesting to see that the hypothesis – even if tested in terms of growth rates – still holds true at a much finer level of sectoral disaggregation (table 4), at the firm level (table 5) and using a different indicator (R&D instead of patents).

As to the direction of causality, the tables give almost no support to Schmookler's unidirectional interpretation ("demand-pull"). Our results are consistent with the hypothesis of mutual dependence of demand and innovation, implying that the relationship between demand and innovation ought to be modelled and estimated using simultaneous equations (see, e.g., Johnston [6, p. 439ff.]).

The argument that demand may favour innovation, and innovation may create extra demand (multiplier/accelerator) may be a message to Keynesian macro-economists. On the one hand, government demand management, besides having the multiplier/accelerator effects well known from the textbooks, may influence the flow of innovations. On the other hand, government policy for promotion of innovation (if successful) may increase aggregate demand. In any case, inclusion of innovation indicators may enhance Keynesian demand analysis. So far, our above results may give a hint for future research.

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